Unit 6 Quiz: Cubic & Cube Root Transformations

Name: ______________________________

Period: ____________ Date: _______________

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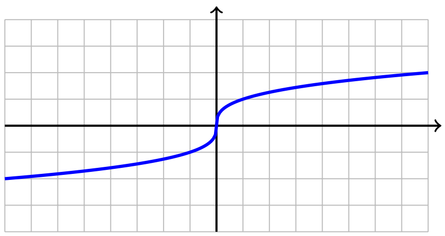
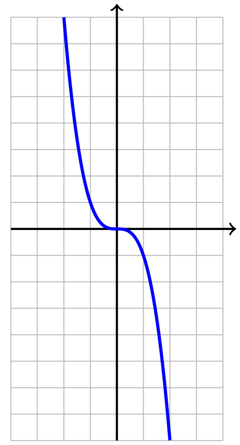
1. Identify whether each graph is a **Cubic (C)** or **Cube Root (CR)**. *(2 points each)*



A. _______



B. _______

2. For the function $f(x) = (x+3)^3 - 2$, identify: *(6 points)*

$a =$ _________ $h =$ _________ $k =$ _________

Center point: _______________

Describe the transformations: ____

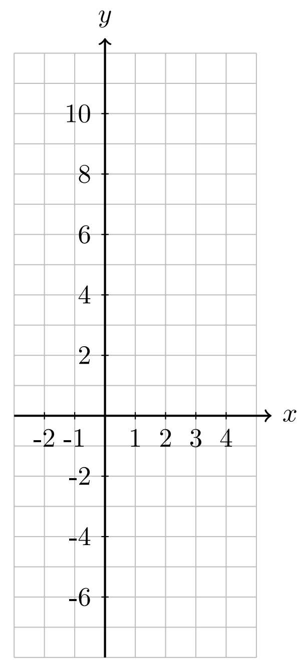
3. For the function $f(x) = -2\sqrt[3]{x}$, identify: *(6 points)*

$a =$ _________ $h =$ _________ $k =$ _________

Center point: _______________

Describe the transformations: ____

4. Graph $f(x) = (x-1)^3 + 2$ *(8 points)*



$a =$ _______ $h =$ _______ $k =$ _______

Center point: (,)

Parent	**Transformed**
$(-2,-8)$	(,)
$(-1,-1)$	(,)
$(0,0)$	(,)
$(1,1)$	(,)
$(2,8)$	(,)

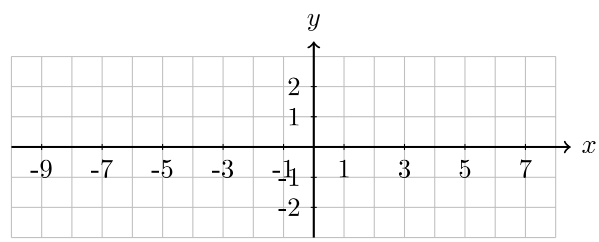
y-intercept: ___________

5. Graph $f(x) = -\sqrt[3]{x+1}$ *(8 points)*

$a =$ _______ $h =$ _______ $k =$ _______

Center point: (,)

Parent	**Transformed**
$(-8,-2)$	(,)
$(-1,-1)$	(,)
$(0,0)$	(,)
$(1,1)$	(,)
$(8,2)$	(,)



y-intercept: ___________

6. Write the equation for a cubic function that is: *(4 points)*

Reflected over the x-axis, shifted left 4, and shifted up 1.

$f(x) =$ ____

Total: ______ / 36 points